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Awake craniotomy with functional mapping for recurrent high-grade IDH-mutant glioma during pregnancy: illustrative case

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BACKGROUND Although gliomas during pregnancy are rare, they present significant clinical and ethical challenges due to the potential effects of pregnancy on tumor behavior and the risks associated with the standard of care. Given the peak incidence of IDH-mutant glioma in the 3rd and 4th decades of life, clinicians face a challenge in balancing maternal oncological outcomes with fetal safety, while navigating the uncertainty surrounding the effects of standard treatment modalities on fetal development.

OBSERVATIONS A 31-year-old woman at 23 weeks of gestation with a history of IDH-mutant glioma presented to the authors' institution with worsening neurological symptoms. MRI indicated tumor progression, and re-resection was performed via awake craniotomy with functional mapping. The maternal-fetal medicine team was consulted to guide the selection of anesthesia and optimize intraoperative positioning, ensuring the safety of both the patient and fetus. Postoperative recovery was uneventful, with stable maternal and fetal outcomes. Pathological analysis confirmed progression to a grade 4 astrocytoma.

LESSONS This case highlights the feasibility of awake craniotomy with functional mapping during pregnancy. Management of gliomas in pregnancy requires a multidisciplinary approach, balancing maternal health and fetal safety, with careful consideration of treatment timing and anesthetic strategies.

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KEYWORDS case report; awake craniotomy; brain mapping; IDH glioma; astrocytoma; pregnancy

Gliomas are the most common primary malignant brain tumors, with incidence closely overlapping with peak fertility. Yet, their occurrence during pregnancy is rare, presenting profound clinical and ethical challenges to the safe delivery of neuro-oncological care. The impact of pregnancy on glioma behavior is poorly understood, and there are currently no clear guidelines regarding the management of the disease at this setting in many pregnant patients' lives.[1] Improved survival among individuals with IDH-mutant gliomas has led to a growing number of patients choosing to pursue pregnancy after diagnosis. Consequently, clinicians are increasingly confronted with the challenge of optimizing maternal oncological outcomes while safeguarding fetal development.[2,3]

Pregnancy complicates glioma management at every stage of care. Standard diagnostic and therapeutic modalities, including gadolinium-enhanced MRI, radiotherapy, chemotherapy, and antiepileptic drugs, carry potential teratogenic or fetal developmental risks.[1,4–6] Recently introduced targeted therapies, such as IDH inhibitors (vorasidenib, ivosidenib), have demonstrated meaningful improvements in progression-free survival, yet their effects on the developing fetus remain unknown, and treatment is typically withheld during gestation.[7–11] As a result, treatment initiation is often delayed or modified, especially in the case of high-grade gliomas, in which a previous study found that standard of care and adjuvant treatment were delayed or denied in 90% of pregnant patients with gliomas.[1]

Beyond therapeutic limitations, pregnancy itself may alter glioma behavior. Multiple cohort studies demonstrate accelerated tumor growth rates, with velocity of diametric expansion increasing when compared to pre-pregnancy rates in nearly 90% of patients.[12–14] Moreover, women with gliomas who were pregnant were found to have increased seizure frequency and neurological deficits compared to their nonpregnant counterparts.[13] In this case report, we discuss the presentation and management of a young woman with a previously

ABBREVIATIONS IHC = immunohistochemical; MFM = maternal-fetal medicine.
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diagnosed IDH-mutant astrocytoma who experienced radiographic and clinical progression during her second trimester of pregnancy. We describe her multidisciplinary management, including awake brain mapping and resection, and review current evidence on the management of gliomas in pregnancy, with particular attention to surgical timing, use of adjuvant therapies, and maternal-fetal considerations in the patient's care.

Illustrative Case

A 31-year-old right-handed woman at 23 weeks 0 days of gestation, with a past medical history notable for gestational diabetes, presented with progressive headaches, word-finding difficulty, and right arm paresthesias for 7 days. She was initially diagnosed 7 years earlier with a left frontal diffuse astrocytoma, IDH1 R132H mutant, ATRX loss, and p53 positive, resected subtotally via awake craniotomy due to proximity to Broca's area (Fig. 1A–F). Histopathological grading at that time revealed a grade 2 tumor, with foci of microvascular proliferation and a Ki-67 index of 10% suggesting a higher grade. She began chemoradiation therapy 4 weeks following resection, with 60 Gy of external beam radiation therapy in 33 fractions and concurrent temozolomide, followed by adjuvant temozolomide. Adjuvant temozolomide was administered for 9 cycles over 11 months. Surveillance brain MRI studies, performed without contrast due to contrast allergy, were stable until 3 years before the most recent presentation. At that point, MRI revealed a small area of tumor recurrence in the surgical bed with slow progression seen on serial MRI studies, for which she was started on ivosidenib 500 mg daily. This was reduced to 250 mg after 9 months of use due to QT interval prolongation. Ivosidenib was continued for an additional 6 months until the patient elected to discontinue therapy in anticipation of pregnancy; she became pregnant 1 year later.

While she remained off systemic treatment during her pregnancy, she was followed with serial noncontrast brain MRI every 3 months. The MRI studies showed tumor stability throughout the first trimester, but at the beginning of the second trimester MRI demonstrated new T2/FLAIR hyperintensity adjacent to the surgical cavity (Fig. 1G and H). Her symptoms worsened, including progressive bifrontal headaches with associated phonophobia and photophobia, as well as word-finding difficulties. Multidisciplinary evaluation and discussion determined that re-resection was indicated.

Preoperative tractography and functional MRI confirmed proximity of the lesion to Broca's area and adjacent speech motor areas. The maternal-fetal medicine (MFM) team was consulted to ensure consideration of both maternal and fetal safety during the procedure. Intraoperative positioning was discussed with the MFM team, and they were present in the operating room for placing the patient into a lateral tilt position with a left shoulder roll. Continuous fetal monitoring was deferred given the periviable gestation age of 23 weeks and 0 days, with pre- and postoperative fetal heart rate assessments performed by the MFM team instead.

The anesthesia for this procedure was performed with an asleep-awake-asleep technique with a laryngeal mask airway. Induction was achieved with propofol, fentanyl, and lidocaine. Sevoflurane along with intermittent infusions of remifentanil and dexmedetomidine was used for anesthesia maintenance. All anesthetics were stopped for the awake portion of the procedure.

An awake craniotomy was performed through the prior incision. Dexamethasone and levetiracetam were administered preoperatively. The cortex immediately overlying the area most concerning for radiographic progression, the inferior frontal gyrus, was then mapped for language function and a safe cortical window was identified. Language function was tested throughout the procedure and remained stable. The center of the tumor was debulked; pial planes were established along the circular sulcus of the insula. Once maximal safe resection was achieved, an intraoperative MRI study was performed, which showed a small amount of residual tumor along the posterior margins of the resection cavity. This additional tissue was removed; it was felt that some areas of FLAIR hypersensitivity posed too great of a language risk for further resection (Fig. 1I and J).

Pathological analysis demonstrated a grade 4 IDH-mutant astrocytoma with a Ki-67 index up to 30%, multiple mitotic figures, and foci of microvascular proliferation (Fig. 2A, B, and E). DNA methylation profiling confirmed a diagnosis of high-grade astrocytoma, IDH mutant. Pyrosequencing for MGMT hypermethylation was negative. Next-generation sequencing and immunohistochemical (IHC) analysis confirmed the presence of IDH R132H mutation (Fig. 2C). IHC analysis also showed aberrant p53 expression and loss of ATRX (Fig. 2D and F).

Postoperatively, the patient experienced transient word-finding difficulty but recovered language function by postoperative day 2. Fetal heart rate evaluation with Doppler remained normal, and she was discharged on postoperative day 3 with stable neurological and obstetric status.

Following resection, due to the time-sensitive nature of treatment, an interdisciplinary team composed of MFM, medical oncology, and radiation oncology personnel recommended early delivery at 34 weeks to expedite chemoradiation therapy for the patient. Cesarean section was recommended and performed at 34 weeks 1 day of gestation. Spinal anesthesia was administered with intrathecal bupivacaine, morphine, and fentanyl injections. The delivery was uncomplicated, and she delivered a healthy 2700-g neonate.

Eighteen days following delivery, the patient began concurrent radiotherapy (59.4 Gy) and daily temozolomide (75 mg/m^2) over a 6-week course. At 6 months postpartum, the patient continues adjuvant temozolomide and reports good overall health. There have been no delays in administration of treatment. The patient reports some adverse effects immediately following each cycle of temozolomide, including fatigue, headaches, and nausea, which is managed with 4 mg of ondansetron as needed. The neonate has experienced no complications and is meeting developmental milestones for age.

A graphical description of the patient's clinical course is outlined in Fig 3.

Informed Consent

The necessary informed consent was obtained in this study.

Discussion

Observations

Gliomas diagnosed before or during pregnancy demand individualized, multidisciplinary care that weighs maternal oncological control against fetal safety. Evidence remains limited to retrospective cohorts and case series, with no prospective trials to define standards of care. Nevertheless, several principles emerge regarding surgical timing, anesthetic strategy, radiotherapy feasibility, systemic therapy, and perioperative medications, all of which should be contextualized by gestational age, tumor biology, and the mother's neurological trajectory.

Surgery in Pregnancy

Resection remains the cornerstone of glioma management. Awake craniotomy with functional mapping has been successfully performed

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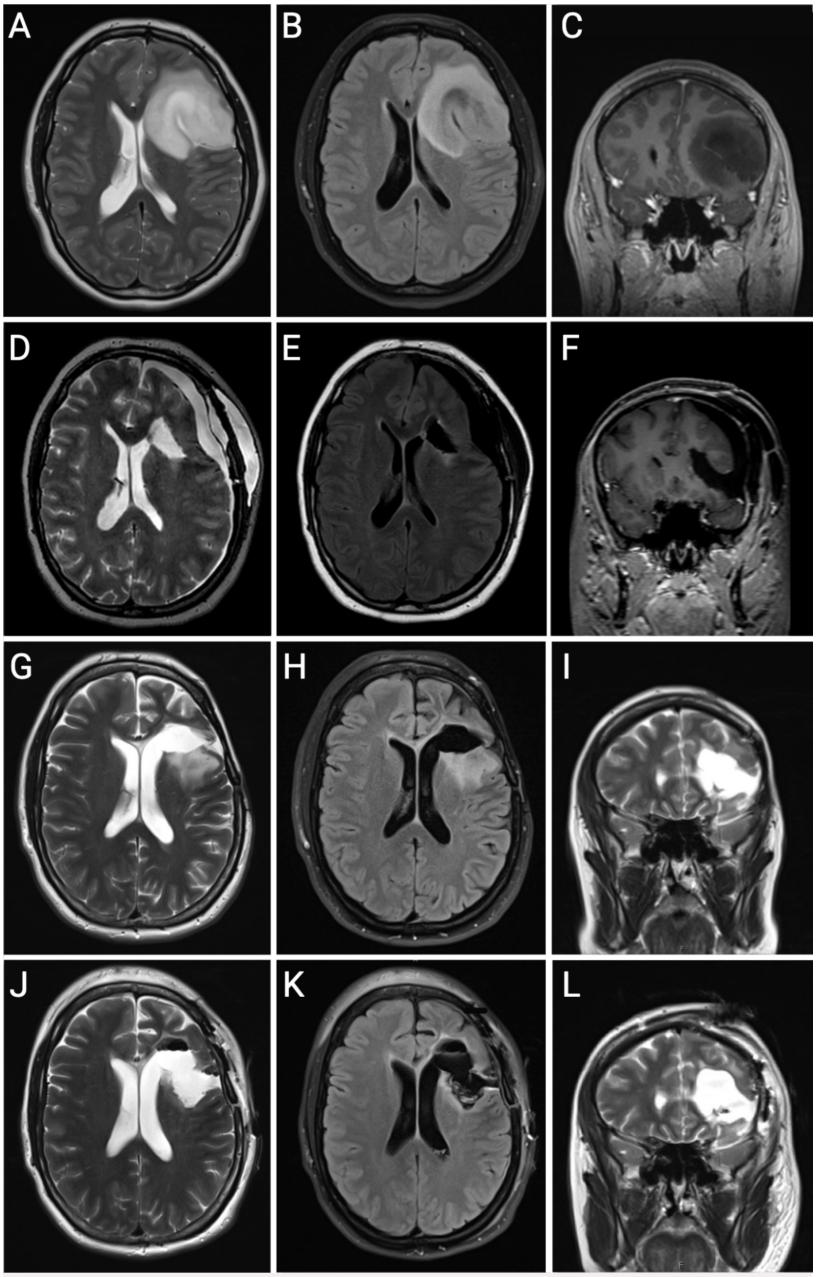


FIG. 1. Preoperative and postoperative MR images of the primary resection and secondary resection 7 years later. Preoperative axial T2-weighted (**A**), axial FLAIR (**B**), and coronal T1-weighted (**C**) images demonstrating the lesion at the time of diagnosis. Postoperative axial T2-weighted (**D**), axial FLAIR (**E**), and coronal T1-weighted (**F**) images showing the original resection cavity. Preoperative axial T2-weighted (**G**), axial FLAIR (**H**), and coronal T2-weighted (**I**) images showing the previous resection cavity and adjacent area of enhancement. Postoperative axial T2-weighted (**J**), axial FLAIR (**K**), and coronal T2-weighted (**L**) sequences demonstrating the resection cavity after repeat resection. Pre- and postoperative T2-weighted coronal sequences were not available from the first resection. Created in BioRender.

during pregnancy and can enable maximal safe resection adjacent to language and motor cortices.[15–18] The second trimester is generally preferred: it avoids first-trimester exposure during organogenesis and increased miscarriage risk, and precedes the obstetric and anesthetic complexities of the third trimester (airway edema, aortocaval compression, preterm labor).[14,16]

Preoperative coordination with the MFM team is essential to optimize both maternal and fetal safety. Intraoperative positioning must

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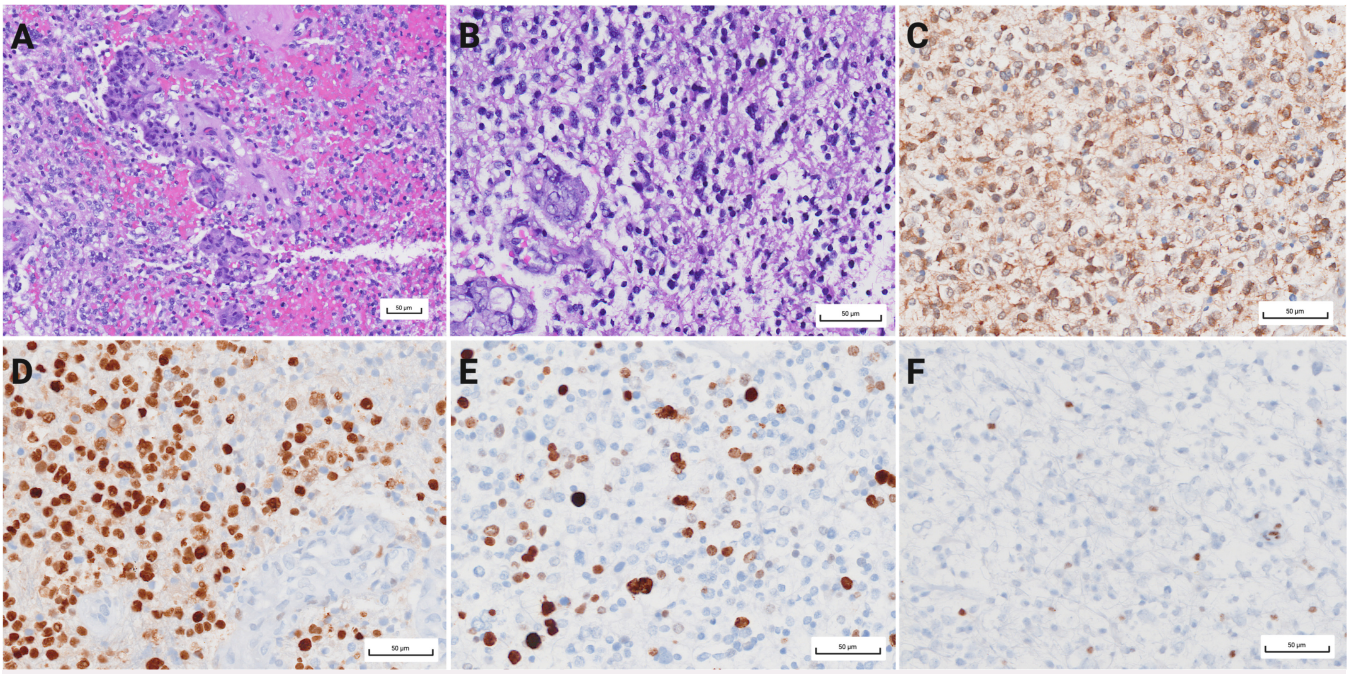


FIG. 2. Histological slides demonstrating diagnosis of high-grade IDH-mutant glioma. **A:** H&E stain showing prominent vascular proliferation. **B:** H&E stain showing a cellular proliferation of atypical tumor cells with hyperchromatic pleomorphic nuclei and clear to eosinophilic cytoplasm. **C:** IDH R132H IHC stain illustrating retained nuclear staining. **D:** A p53 IHC stain demonstrating aberrant nuclear staining. **E:** Ki-67 IHC stain showing increased proliferative activity. **F:** ATRX IHC stain showing loss of expression. Original magnification ×20 (A), ×40 (B–F). Created in BioRender.

account for uteroplacental perfusion, while also accommodating neurosurgical exposure. Fetal monitoring strategies should be individualized based on gestational age, with most centers recommending pre- and postoperative assessments in the second trimester, with continuous intraoperative monitoring a consideration for later gestation age, although this decision is individualized based on the clinical scenario, feasibility of monitoring and obstetric interventions, and patient preference. When indicated, surgery should not be delayed, as maternal neurological decline itself poses substantial risk to the pregnancy dyad. Just as importantly, risks and benefits should be discussed in depth with the patient, ensuring that surgical timing, anesthetic considerations, and maternal-fetal trade-offs are incorporated into shared decision-making.

In this case, it was decided to pursue resection at 23 weeks due to the sudden onset of constitutional symptoms, including headaches and word-finding difficulties, paired with the comparatively favorable timing of surgery in the second trimester of gestation.[14,16] While the effects of pregnancy on glioma biology are poorly understood, it has been hypothesized that hormones released during gestation may hasten tumor growth, resulting in disease progression more rapidly than otherwise expected.[12] Pregnancy has been shown to decrease progression-free survival, supporting this hypothesis.[19] In addition, pregnancy is associated with an increased risk of severe glioma-related symptoms, such as seizures.[12,13] Via an interdisciplinary discussion of the risks and benefits of resection with the patient, it was decided that a second-trimester resection was the most favorable course of action

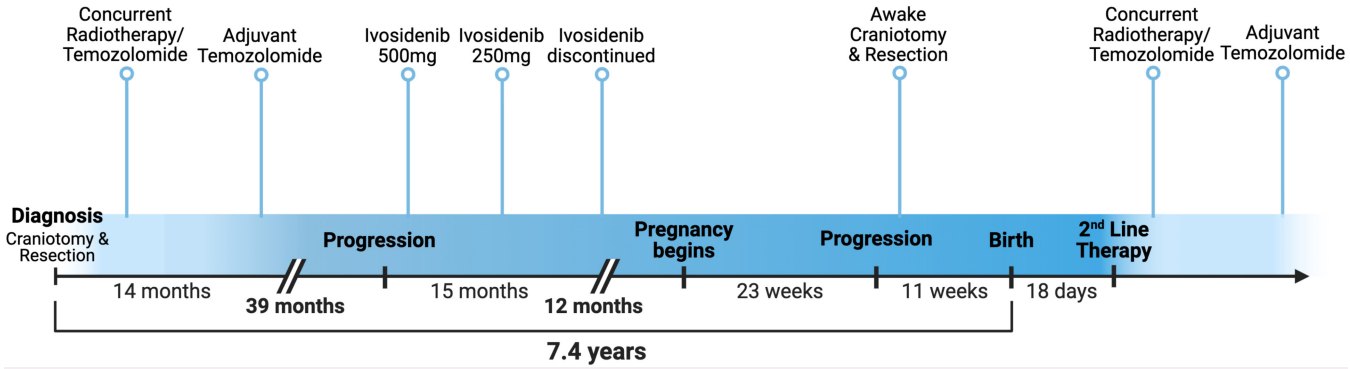


FIG. 3. Graphical representation of the patient's clinical course from initial presentation to completion of pregnancy. Not to scale. Created in BioRender.

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to preserve the health of the pregnancy dyad, enabling continuation of the pregnancy while relieving the patient's worsening symptoms.

Anesthesia During Pregnancy

Modern awake techniques can be tailored to pregnancy. Propofol has been reported safe for the induction of sedation in awake craniotomies for pregnant patients, but prolonged use is not advised.[20] Both remifentanil and dexmedetomidine are used regularly in anesthetic management of pregnant patients, during awake craniectomy as well as other surgical procedures.[17,21] While remifentanil crosses the placental barrier, it is rapidly cleared by the fetus, conferring minimal risk; it is commonly used in fetal monitoring procedures.[22] Dexmedetomidine has been similarly reported safe for use in pregnant patients due to decreased placental transfer to the fetus, resulting in relatively low fetal concentrations.[17,21,23,24] Sevoflurane is also not known to cause harm to the fetus.[25]

Radiation Therapy in Pregnancy

Cranial radiotherapy is often deferred until postpartum, particularly in early gestation.[14,26] Nevertheless, fetal exposure during cranial radiation therapy derives almost entirely from scatter; with planning, the fetal dose can be kept very low. Dosimetric studies suggest that fetal doses of ≤ 10 cGy are unlikely to increase the risk of growth restriction, malformation, or miscarriage, whereas doses > 50 cGy carry risks across trimesters.[14,27,28] Techniques to minimize dose include meticulous shielding of the abdomen/pelvis, use of beams < 10 mV, and optimal linac choice to avoid radiation leakage.[26] Stereotactic radiosurgery (e.g., CyberKnife) in the late second/third trimester has been reported with fetal doses < 10 cGy and without adverse outcomes in carefully selected cases.[29] When faced with aggressive progression that threatens maternal status, timely radiotherapy may be warranted, particularly later in gestation.[30]

Chemotherapy in Pregnancy

While it is generally accepted that chemotherapeutic agents should be avoided during the first trimester due to the risk of teratogenicity, after this point chemotherapy can be considered on an individualized basis depending on cancer type, stage, chemotherapy agent, and patient preferences. While the risk of major malformations is minimal, there may remain ongoing risks to the pregnancy to consider, including preterm birth and impaired fetal growth.Alkylating agents, including temozolomide, which are frequently utilized in the treatment of high-grade gliomas, have some of the highest rates of adverse pregnancy outcomes and are able to cross the placenta.[5,27] However, one case series of patients with glioma who became pregnant during alkylating chemotherapy treatment reported no birth defects; however, chemotherapy was discontinued once the pregnancies were discovered.[13,27] In practice, temozolomide is generally deferred until postpartum; if treatment is unavoidable, later-trimester use after detailed counseling may be considered on a case-by-case basis.[31] The effects of IDH inhibitors (ivosidenib, vorasidenib) on embryofetal development remain insufficiently characterized; current recommendations are to withhold during pregnancy.[7–11]

Antiepileptic Drugs and Steroids in Pregnancy

Seizures endanger both the mother and fetus; seizure prevention outweighs potential medication risks. Lamotrigine and levetiracetam have the most favorable reproductive safety profiles among commonly used agents and are preferred, ideally as monotherapy with folic acid supplementation.[4,32] Clearance for both lamotrigine and levetiracetam increases during pregnancy, necessitating increased dosing throughout trimesters and close monitoring of drug serum levels.[33–35]

Corticosteroids remain essential for the management of vasogenic edema; in addition to maternal neurological benefit, some antenatal corticosteroids promote pulmonary surfactant production in the fetus, allowing for safer early delivery if necessary.[36]

Effects of Pregnancy on Glioma Biology

A central and often challenging component of counseling in these cases is whether continuation of pregnancy adversely affects maternal oncological outcomes. Pregnancy is known to have profound effects on glioma biology. Pregnancy has been associated with not only acceleration of tumor growth rates, but also increased seizure frequency.[12,13] Hormones released during pregnancy, including progesterone and insulin-like growth factors, may directly stimulate glioma growth, although this effect appears to be more pronounced in hormone-sensitive tumors.[12,37] Pregnancy has been shown to significantly decrease progression-free survival in patients with IDH-mutant gliomas.[19] However, while up to 90% of patients experience imaging progression during pregnancy, pregnancy itself has not been associated with decreased overall survival.[13,14]

In clinical practice, management decisions must balance these uncertainties with the risks of delaying oncological therapy. Contemporary literature and consensus guidelines emphasize that care should be guided by a values-based informed consent process.[37,38] The American Society of Clinical Oncology guidelines for the management of cancer during pregnancy state that treatment decisions should be individualized in discussion with the patient, taking into account the potential harms of delaying necessary maternal treatment while balancing fetal risk.[38] Accordingly, continuation of pregnancy, early delivery, or pregnancy termination may be considered depending on maternal diagnosis, progression of disease, or urgency of treatment.[38] Ultimately, these decisions are highly individualized and require multidisciplinary input, with careful attention to patient autonomy, beneficence, and shared decision-making.[37]

Lessons

This case illustrates the feasibility and safety of performing awake craniotomy with functional mapping for glioma resection during the second trimester of pregnancy. During pregnancy, surgical care becomes particularly important, as fetal exposure to chemotherapy and radiation therapy carries substantially higher risks than those associated with resection. Although evidence remains limited to small series, available data and our experience support surgery when maternal neurological function is threatened, as deterioration poses the greatest risk to both the mother and fetus. Radiation therapy and chemotherapy should generally be deferred until after delivery, unless maternal disease progression mandates earlier treatment. When radiotherapy is unavoidable, careful dosimetric planning and fetal shielding can limit fetal exposure to below the 10-cGy threshold considered low risk. Alkylating agents, including temozolomide, and novel IDH inhibitors should be withheld during pregnancy when possible. Among perioperative medications, dexamethasone and antiseizure drugs such as levetiracetam remain preferred for their relative fetal safety. The main limitation of this case report is the absence of long-term follow-up data, as the patient is currently 6 months postpartum. Ultimately, management should be individualized, multidisciplinary, and centered on preserving maternal health as well as safeguarding fetal outcomes.

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Disclosures
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Author Contributions
Conception and design: Orringer, Fang, Weiss. Acquisition of data: Orringer, Fang. Analysis and interpretation of data: Orringer, Fang, Bhanja. Drafting the article: Orringer, Fang, Weiss, Bhanja. Critically revising the article: all authors. Reviewed submitted version of manuscript: Orringer, Fang, Bhanja, Penfield, Ard. Approved the final version of the manuscript on behalf of all authors: Orringer. Study supervision: Orringer.

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